20 Şubat 2024 Salı 11:09

Ex $f(\overbrace{x_1, x_2, x_3}^{x}) = 3x_1x_2 + x_2^2 + x_3^2$

$f: \mathbb{R}^3 \longrightarrow \mathbb{R}$

$(x_1, x_2, x_3) \longmapsto 3x_1x_2 + x_2^2 + x_3^2$

$$\nabla f(x) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \partial f/\partial x_3 \end{bmatrix} = \begin{bmatrix} 3x_2 \\ 3x_1 + 2x_2 \\ 2x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_2 = 0 \quad x_1 = 0 \quad x_3 = 0$

$\boxed{x^* = (0,0,0)}$ → stationary point

$$\nabla^2 f(x) = \begin{bmatrix} 0 & 3 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$\nabla^2 f(x^*) = \checkmark$

$|\nabla^2 f(x) - \lambda I|$ → determinant.

$$\lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{vmatrix} -\lambda & 3 & 0 \\ 3 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} \quad 3\times 3$$

$$= (2-\lambda)\begin{vmatrix} -\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = (2-\lambda)\left[(-\lambda)(2-\lambda) - 9\right]$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad ad - bc$$

$$= (2-\lambda)\left[-2\lambda + \lambda^2 - 9\right]$$

$$= (2-\lambda)(\lambda^2 - 2\lambda - 9)$$

$ax^2 + bx + c = 0$

$\lambda = 2$

$\Delta = 4 - 4\cdot(-9) > 0$

$4 + 36 = 40$

$\lambda = \dfrac{-b \mp \sqrt{\Delta}}{2a}$

$\lambda_{1,2} = \dfrac{2 \mp \sqrt{40}}{2}$

$= 1 \mp \sqrt{10}$

$1 - \sqrt{10} < 0$

$\lambda_2 = 1 + \sqrt{10} > 0$

$\lambda_3 = 1 - \sqrt{10} < 0$

$\Rightarrow \nabla^2 f(x^*)$ is indefinite matrix

→ $(0,0,0)$

$\Rightarrow$ saddle pt

If $\nabla^2 f(x)$

$$\begin{bmatrix} 3x_1x_2 & 0 & 3 \\ & 5x_2x_3^2 & \\ - & - & - \end{bmatrix}$$

evaluate substitute → x^* → stationary pt

$\nabla^2 f(x^*)$

↓ next step

Find eigenvalue

$|\nabla^2 f(x^*) - \lambda I| = 0$

Ex Find all extremum pts for $f(x_1, x_2) = x_1^2 x_2 - x_2^2 x_1$

Step 1 Find stationary pts $\nabla f(x_1, x_2) = 0$ Solve for $x_1, x_2 = ?$

$$\nabla f(x_1, x_2) = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{bmatrix} = \begin{bmatrix} 2x_1x_2 - x_2^2 \\ x_1^2 - 2x_1x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$2x_1x_2 - x_2^2 = 0 \Rightarrow x_2(2x_1 - x_2) = 0 \Rightarrow x_2 = 0$ or $2x_1 = x_2$

$x_1^2 - 2x_1x_2 = 0 \Rightarrow x_1(x_1 - 2x_2) = 0 \Rightarrow x_1 = 0$ or $x_1 = 2x_2$

if $x_2 = 0 \Rightarrow x_1 = 0$ $\boxed{(0,0)}$

$\frac{1}{4}\left(2\cdot\frac{1}{2} - \frac{1}{4}\right) \neq 0$

if $x_1 = 0 \Rightarrow x_2 = 0$

$\frac{1}{4}\left(2\cdot\frac{1}{2} - \frac{1}{4}\right) \neq 0$

if $2x_1 = x_2$ ← $x_1 = 2x_2$

$4x_2^2 - x_2 = 0$

$x_2(4x_2 - 1) = 0$

$x_2 = 0$ $\quad x_2 = 1/4$ $\quad x_1 = \frac{1}{2}$

$x_2 = \frac{1}{4} \rightarrow x_1 = 1/2$

Step 2 $\nabla^2 f(0,0) = ?$

$$\nabla^2 f(x_1, x_2)\Big|_{(0,0)} = \begin{bmatrix} 2x_2 & 2x_1 - 2x_2 \\ 2x_1 - 2x_2 & -2x_1 \end{bmatrix}\Bigg|_{x = (0,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$(0,0)$ is degenerate point since $|\nabla^2 f(0,0)| = 0$